

Expressions

Introduction to Programming

StayAhead Training; Sep, 2024

Expression

Expressions

- **Expression:** statement that yields a new value, e.g.:
 - $x + y$
 - $x > 0$
 - `strip(name)`

Expression

Expressions

- Pseudocode, e.g.:
 - `COMPUTE Sum = Num1 + Num2`
- Or, simply:
 - `SET Sum to Num1 + Num2`

Expression

Expressions

- PY: result = num * 4
- -----
- JS: var result = num * 4;
- -----
- JV: int result = num * 4;
- -----
- C#: int result = num * 4;

Operator

Expressions

- **Operator:** symbol/keyword that instructs the computer to do something, e.g.:
 - =
 - +
 - >
 - in

Operator

Expressions

- PY: result = num * 4
- -----
- JS: var result = num * 4;
- -----
- JV: int result = num * 4;
- -----
- C#: int result = num * 4;

Operand

Expressions

- **Operand:** the subject of the operator's action
- The things either side of the operator
- May be variables or literals

Operand

Expressions

- PY: result = num * 4
- -----
- JS: var result = num * 4;
- -----
- JV: int result = num * 4;
- -----
- C#: int result = num * 4;

Order of Execution

Expressions

- The expression is evaluated first:
 - `num * 4`
- The result of which is then assigned to the variable:
 - `result = 64` (assuming `num` is 16)
- The assignment always happens last

Arithmetic Operators

Expressions

	Python	JavaScript	Java	C#
Addition	+	+	+	+
Subtraction	-	-	-	-
Multiplication	*	*	*	*
Division	/	/	/	/
Modulus (remainder)	%	%	%	%

Division with Unexpected Results

Expressions

- PY: `result = 9 / 4`
- `-----`
- JS: `var result = 9 / 4;`
- `-----`
- JV: `float result = 9 / 4;`
- `-----`
- C#: `float result = 9 / 4;`

Casting

Expressions

- PY: `result = 9 / float(4)`
- -----
- JS: `var result = 9 / parseFloat(4);`
- -----
- JV: `float result = 9 / (float) 4;`
- -----
- C#: `float result = 9 / (float) 4;`

Casting

Expressions

- PY: `result = int(2.7)`
- -----
- JS: `var result = parseInt(2.7);`
- -----
- JV: `int result = (int) 2.7f;`
- -----
- C#: `int result = (int) 2.7f;`

Relational Operators

Expressions

	Python	JavaScript	Java	C#
Equal to	==	==	==	==
Not equal to	!=	!=	!=	!=
Greater than	>	>	>	>
Greater than or equal to	>=	>=	>=	>=
Less than	<	<	<	<
Less than or equal to	<=	<=	<=	<=

Relational Operators

Expressions

- PY: `result = x >= y`
- _____
- JS: `var result = x >= y;`
- _____
- JV: `boolean result = x >= y;`
- _____
- C#: `bool result = x >= y;`

Logical Operators

Expressions

	Python	JavaScript	Java	C#
AND	and	&&	&&	&&
OR	or			
XOR (eXclusive OR)	^	^	^	^

Logical Operators

Expressions

- PY: `can_purchase = has_camera == True and price <= 100`
- -----
- JS: `var canPurchase = hasCamera == true && price <= 100;`
- -----
- JV: `boolean canPurchase = hasCamera == true && price <= 100;`
- -----
- C#: `bool canPurchase = hasCamera == true && price <= 100;`

Logical Operator Rules

Expressions

Operator	Rule
AND	The result is true if both of the operands evaluates to true
OR	The result is true if one of the operands evaluates to true
XOR	...if one of the operands evaluates to true, and the other evaluates to false

Truth Tables

Expressions

Operand #1	Operand #2	AND	OR	XOR
True	True	True	True	False
True	False	False	True	True
False	True	False	True	True
False	False	False	False	False

Evaluating Boolean Variables

Expressions

- PY: `result = has_camera == True`
- _____
- JS: `var result = hasCamera == true;`
- _____
- JV: `boolean result = hasCamera == true;`
- _____
- C#: `bool result = hasCamera == true;`

Evaluating Boolean Variables

Expressions

- PY: `result = has_camera`
- _____
- JS: `var result = hasCamera;`
- _____
- JV: `boolean result = hasCamera;`
- _____
- C#: `bool result = hasCamera;`

Evaluating Boolean Variables

Expressions

- PY: `can_purchase = has_camera and price <= 100`
- -----
- JS: `var canPurchase = hasCamera && price <= 100;`
- -----
- JV: `boolean canPurchase = hasCamera && price <= 100;`
- -----
- C#: `bool canPurchase = hasCamera && price <= 100;`

The NOT Operator

Expressions

	Python	JavaScript	Java	C#
NOT	not	!	!	!

The NOT Operator

Expressions

- PY: `result = not x`
- _____
- JS: `var result = !x;`
- _____
- JV: `boolean result = !x;`
- _____
- C#: `bool result = !x;`

The NOT Operator

Expressions

- PY: `can_purchase = not has_camera and price <= 100`
- -----
- JS: `var canPurchase = !hasCamera && price <= 100;`
- -----
- JV: `boolean canPurchase = !hasCamera && price <= 100;`
- -----
- C#: `bool canPurchase = !hasCamera && price <= 100;`

Assignment Operators

Expressions

	Python	JavaScript	Java	C#
Assignment	=	=	=	=
Addition asgnmnt	+=	+=	+=	+=
Subtraction asgnmnt	-=	-=	-=	-=
Multiplication asgnmnt	*=	*=	*=	*=
Division asgnmnt	/=	/=	/=	/=
Modulus asgnmnt	%=	%=	%=	%=

Assignment Operators

Expressions

- PY: $x = x + 2$
- $-----$
- JS: $x = x + 2;$
- $-----$
- JV: $x = x + 2;$
- $-----$
- C#: $x = x + 2;$

Assignment Operators

Expressions

- PY: `x += 2`
- `_____`
- JS: `x += 2;`
- `_____`
- JV: `x += 2;`
- `_____`
- C#: `x += 2;`

Precedence

Expressions

- PY: `result = 1 + 2 * 3`
- `-----`
- JS: `var result = 1 + 2 * 3;`
- `-----`
- JV: `int result = 1 + 2 * 3;`
- `-----`
- C#: `int result = 1 + 2 * 3;`

Precedence

Expressions

- Multiplication/division takes precedence
 - $2 * 3$
- ...over addition/subtraction
 - $1 + 6$

Precedence (BO-DM-AS)

Expressions

- PY: `result = (1 + 2) * 3`
- _____
- JS: `var result = (1 + 2) * 3;`
- _____
- JV: `int result = (1 + 2) * 3;`
- _____
- C#: `int result = (1 + 2) * 3;`

Precedence (None)

Expressions

- PY: `result = 6 / 3 * 4`
- `-----`
- JS: `var result = 6 / 3 * 4;`
- `-----`
- JV: `int result = 6 / 3 * 4;`
- `-----`
- C#: `int result = 6 / 3 * 4;`